

CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

Chapter 5: Tales by Dots and Lines

Topic-wise short notes with solved examples

The mean as a balance point, how the mean and median change when data change, the mean and median of data given with frequencies, spreadsheets, and reading and drawing line graphs.

1 Mean and median

- **Mean** (average) = $\frac{\text{sum of all the values}}{\text{number of values}}$.
- **Median**: the middle value when the data are arranged in order. With an even number of values, it is the mean of the two middle values.
- Sum of the values = mean $\times$ number of values. This is used again and again.

Example 1.1 Find the mean and median of 8, 3, 10, 13, 4, 6, 7, 7, 8, 8, 5.

SOLUTION

Sum = 79 and there are 11 values, so the mean is $\frac{79}{11} \approx 7.18$.

In order: 3, 4, 5, 6, 7, 7, 8, 8, 8, 10, 13. The 6th value is 7.

Mean about 7.18; median 7

2 The mean as a balance point

- On a dot plot, the **total distance** of the values below the mean from the mean equals the total distance of the values above it. The mean is the point where the data balance.
- The mean is not always midway between the smallest and largest values.
- There is only one such balance point.



Data 10, 10, 11, 17: distances on the left $2 + 2 + 1 = 5$, on the right 5

Example 2.1 Check the balance property for the data 4, 6, 7, 11, 12.

SOLUTION

$$\text{Mean} = \frac{40}{5} = 8.$$

Below the mean: $4 + 2 + 1 = 7$. Above the mean: $3 + 4 = 7$.

The distances balance

3 Adding or removing values

- Including a value **greater than the mean** increases the mean. Including a value **less than the mean** decreases it. Including a value **equal to the mean** leaves it unchanged.
- Removing a value has the opposite effect: removing a value above the mean lowers the mean.
- Several values can be added without changing the mean if their distances above and below the mean balance, for example $m - 3$ and $m + 3$, or $m - 1$, $m - 2$ and $m + 3$.
- The **median** behaves in a similar way: including a value greater than the median cannot lower it, and including a value smaller than the median cannot raise it. The median may also stay the same.

Example 3.1 The mean of 9 numbers is 20. A tenth number, 40, is included. Find the new mean.

SOLUTION

$$\text{New sum} = 180 + 40 = 220$$

22

Example 3.2 The data 3, 5, 8, 8, 9, 12, 14 have median 8. What is the median after 11 is included?

SOLUTION

In order: 3, 5, 8, 8, 9, 11, 12, 14. The two middle values are 8 and 9.

8.5

4 Changing every value

- If the same number k is **added** to every value, the mean increases by k . If k is subtracted, the mean decreases by k .
- If every value is **multiplied** by k , the mean is multiplied by k .
- Reason:
$$\frac{(x_1 + k) + (x_2 + k) + \dots + (x_n + k)}{n} = \frac{x_1 + x_2 + \dots + x_n}{n} + \frac{nk}{n} = a + k.$$
- The median changes in exactly the same way.

Example 4.1 The mean height of 24 students, measured with shoes on, is 150.2 cm. The shoes add 1 cm. Find the correct mean height.

SOLUTION

Every value is 1 cm too large, so the mean is 1 cm too large.

149.2 cm

Example 4.2 The mean of a data set is 12. Every value is doubled and then 5 is added. Find the new mean.

SOLUTION

$$2 \times 12 + 5$$

29

5 Finding an unknown value

- Use: $\text{sum} = \text{mean} \times \text{number of values}$.

Example 5.1 The mean weight of 10 players is 39.2 kg. Nine of the weights add up to 349 kg. Find the tenth.

SOLUTION

$$\text{Total} = 39.2 \times 10 = 392$$

43 kg

Example 5.2 The mean of 15 values is 25.6. One value was recorded as 3 more than it should be. Find the correct mean.

SOLUTION

$$\text{Recorded sum} = 25.6 \times 15 = 384. \text{ Correct sum} = 381.$$

$$\frac{381}{15} = 25.4$$

Example 5.3 The mean of 8, 13, 10, 4, 5, 20, y , 10 is 10.375. Find y .

SOLUTION

$$\text{Sum} = 10.375 \times 8 = 83, \text{ and the known values add up to } 70.$$

$$y = 13$$

6 Mean and median with frequencies

- When each value occurs several times, multiply each value by its **frequency** before adding.

$$\text{Mean} = \frac{\text{sum of (value} \times \text{frequency)}}{\text{sum of the frequencies}}$$

- Do **not** just average the different values.
- **Median**: add the frequencies one after another (running totals) until you reach the middle position(s).

Family size	3	4	5	6	7	Total
Number of students	4	10	8	5	3	30
Size × number	12	40	40	30	21	143

Example 6.1 Find the mean and median family size from the table above.

SOLUTION

$$\text{Mean} = \frac{143}{30} \approx 4.77.$$

There are 30 values, so the median is the mean of the 15th and 16th values. Running totals: 4, 14, 22. Positions 15 to 22 all hold the value 5.

Mean about 4.77; median 5

7 Spreadsheets

- A **spreadsheet** is a table of **cells** on a computer. Columns are labelled A, B, C, ... and rows 1, 2, 3, ... The cell in column E and row 5 is **E5**.
- A cell can hold text, a number or a **formula**. A formula begins with =.
- A row or column of cells is written Start:End. B3:G3 means the cells from B3 to G3; D2:D6 means the cells from D2 to D6.

Formula	What it gives
=SUM(B3:G3)	The total of cells B3 to G3
=AVERAGE(B2:B23)	The mean of cells B2 to B23
=MEDIAN(B2:B23)	The median of cells B2 to B23

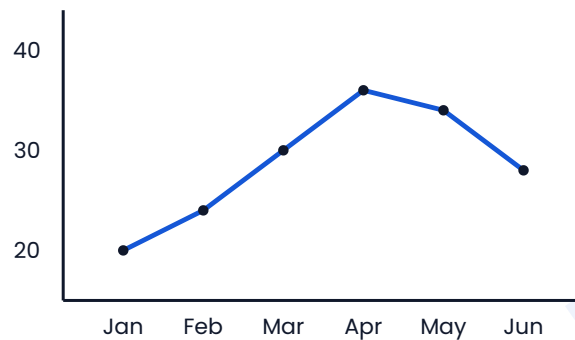
- When a number in the table is changed, every formula that uses it updates automatically.

8 Line graphs

- A **line graph** joins data points by line segments. It is best for showing how a quantity **changes over time**.
- The horizontal axis usually shows time; the vertical axis shows the quantity. Always read the titles, the scale and the key first.
- **Reading a graph in two steps**: (1) identify what is given: axes, scale, units, which line is which; (2) infer and interpret: highest and lowest points, rises and falls, comparisons.
- A **steeper** segment means a faster change. A flat segment means no change.

- Two or more lines on the same axes let us compare trends. With many time points, a line graph is much clearer than a clustered bar graph.
- Ask how the data were collected. This shows what the graph can and cannot tell us.

Month	Jan	Feb	Mar	Apr	May	Jun
Maximum temperature (°C)	20	24	30	36	34	28



Line graph of the monthly maximum temperature (°C)

Example 8.1 From the graph above: (i) in which month is the temperature highest? (ii) Between which two months is the rise greatest? (iii) What is the mean for the six months?

SOLUTION

- (i) The highest point is April, at 36.
 (ii) The steepest rising segments are Feb to Mar and Mar to Apr, each a rise of 6.
 (iii) $\frac{20 + 24 + 30 + 36 + 34 + 28}{6} = \frac{172}{6}$

April; Feb to Mar and Mar to Apr; about 28.7°C

Example 8.2 A line graph shows a shop's visitors over a week: 16, 19, 10, 14, 20, 22, 35 from Monday to Sunday. Describe the trend.

SOLUTION

The lowest point is Wednesday (10). From Wednesday the line rises every day.

Visitors dip mid-week and climb to a peak of 35 on Sunday; the steepest rise is from Saturday to Sunday.

9 Other ways of showing data

- **Infographics** combine pictures, maps, colours and numbers to tell a story quickly. Read the colour scale or key before drawing conclusions.
- An **activity strip** shows how a day is spent: a strip of 48 boxes, one for each half hour, coloured by activity.
- A graph can support only what its data show. "No power cuts" cannot be concluded from a graph of how many homes use electricity for lighting.

10 Common mistakes

- Averaging the distinct values and ignoring their frequencies.
- Finding the median without arranging the data in order.
- Assuming the mean is midway between the smallest and largest values.
- Thinking that adding any new value raises the mean. It depends on whether the value is above or below the mean.
- Reading a line graph without checking the scale on the vertical axis.
- Concluding things about individuals from an average.

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